

The Erosion of Public Capital in Portugal

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Abstract

This article documents the erosion of Portugal’s public capital stock since 2013 and assesses its macroeconomic implications. I extend the IMF capital series through 2027 using European Commission data and construct scenarios through 2035 for alternative infrastructure-project pipelines. In 2026, the stock is projected to remain 12.2 percent below its 2013 peak, while its ratio to GDP is projected to fall from 77.2 percent in 2013 to 52.6 percent. Short-horizon regressions show positive co-movement between public-capital and labor-productivity growth. A calibrated production-function exercise suggests that the capital decline may reduce 2026 GDP per capita by 0.7–5.3 percent, with a central estimate of 1.4 percent. Announced investment projects slow, but do not reverse, the projected decline in the capital-to-GDP ratio.

JEL codes: E22, H50, H54, O47.

Keywords: public investment, public capital, infrastructure, Portugal, fiscal policy.

1 Introduction

Public investment is a small but consequential component of government expenditure. It creates assets that support production and the delivery of public services over many years, including transport networks, schools and universities, hospitals, courts, administrative buildings, defense installations, and digital infrastructure. Because these assets depreciate physically and become technologically obsolete, maintaining the productive capacity of the public sector requires continuing expenditure on replacement and modernization. A sustained gap between gross investment and depreciation eventually reduces the stock of usable public capital.

Portugal provides a clear illustration of this mechanism. Fiscal consolidation after the euro-area sovereign debt crisis placed government expenditure under severe pressure. Adjustment was unavoidable given the financing constraints, but its composition remained a policy choice. Public investment is politically easier to reduce than social transfers or current government consumption because the immediate effects are often less visible. Maintenance can be deferred, projects can be

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postponed, and service quality may initially be preserved by using existing assets more intensively. The budgetary savings appear immediately, whereas the costs accumulate gradually. Leading Portuguese economic-policy institutions, including the Public Finance Council and the Bank of Portugal, have increasingly drawn attention to the persistent weakness of public investment and its implications for the public capital stock ([Conselho das Finanças Públicas, 2026](#); [Banco de Portugal, 2026](#)).

This article extends and updates the analysis in [Faria-e-Castro \(2021\)](#), which documented a sharp decline in Portuguese public investment after 2010 and the onset of a contraction in the estimated public capital stock around 2013. The intervening years make it possible to assess whether that episode was temporary. They also encompass the pandemic and the Recovery and Resilience Plan (Plano de Recuperação e Resiliência, PRR), which provided a substantial, European-funded boost to public investment.

This article makes five contributions. First, it extends the IMF capital-stock series through 2027 using the European Commission’s July 2026 AMECO vintage. Second, it distinguishes the historical record from the exceptional increase in investment forecast for 2026. Third, it quantifies the decline in real public capital and its ratio to GDP, compares investment with current expenditure, and calculates the investment flow required to preserve the 2013 stock. Fourth, it examines the possible macroeconomic implications using descriptive productivity regressions and a calibrated Cobb–Douglas counterfactual that holds public capital at its 2013 level. Fifth, it analyzes how the announced project pipeline, together with a speculative extension that includes the new Lisbon airport, could affect public investment and the public capital stock through 2035 under alternative assumptions about project execution and timing.

The main finding is that the erosion did not reverse during the first half of the 2020s. The public capital stock peaked at about 242.6 bn 2017 international dollars in 2013 and reached a trough of about 210.4 bn in 2024, a cumulative decline of 13.3 percent. Its ratio to GDP fell from 77.2 percent to 53.8 percent over the same period. The July 2026 AMECO forecast implies that public investment will rise to 4.3 percent of GDP in 2026 and that the capital stock will increase more visibly in 2027. Yet the projected 2027 stock remains roughly 9 percent below its 2013 peak. A temporary investment surge is therefore not equivalent to reversing a decade of underinvestment.

The calibrated exercise illustrates why this persistence may matter for macroeconomic performance. Under the central output elasticity of 0.10, the public-capital shortfall relevant for production in 2026 corresponds to output and labor productivity about 1.4 percent below the constant-2013-stock counterfactual. Spread over 2013–2026, this difference is equivalent to an average annual growth drag of approximately 0.11 percentage points. The implied 2026 gap ranges from 0.7 percent under a conservative elasticity of 0.05 to 5.3 percent under the upper estimate of 0.39. The exercise holds other inputs fixed and should be interpreted as a disciplined accounting counterfactual, not as evidence that the decline in public capital causally generated the estimated output gap.

The forward-looking accounting exercise suggests that the announced projects will have a relatively modest effect on the capital stock. In the announced-project scenario, the stock rises through 2030 but resumes its decline thereafter. A more speculative scenario, which includes projects with only preliminary cost estimates and schedules, such as the new Lisbon airport and the Third Tagus Bridge, produces a larger increase through 2035; even then, the projected stock remains below its 2013 peak. These results are sensitive to uncertain assumptions about project execution and

timing, both of which may be affected by administrative capacity constraints.

These calculations are partial-equilibrium accounting exercises and do not attempt to measure the broader macroeconomic or welfare effects of alternative investment scenarios. They measure depreciated acquisition values under an assumed post-2018 depreciation rate, not engineering condition, service quality, social returns, replacement costs, or the human and organizational capital of the state.¹ Section 2 describes the data sources and perpetual-inventory methodology. Section 3 presents the main results for public investment and capital. Section 4 documents simple statistical relationships between public-capital growth and labor-productivity growth and quantifies the potential macroeconomic implications of the post-2013 decline. Section 5 presents the forward-looking project scenarios, and Section 6 concludes.

2 Methodology

2.1 Concepts and data sources

The historical benchmark is the IMF Investment and Capital Stock Dataset, documented in [International Monetary Fund \(2021\)](#). The IMF constructs internationally comparable public and private capital stocks following the perpetual-inventory approach of [Kamps \(2006\)](#) and [Gupta et al. \(2014\)](#). The source workbook used here contains Portugal’s public capital and investment series, together with separately identified PPP capital and investment. The two components are added to form the broad stock and flow used in the update:

$$K_t = K_t^{\text{gov}} + K_t^{\text{PPP}}, \quad (1)$$

$$I_t = I_t^{\text{gov}} + I_t^{\text{PPP}}. \quad (2)$$

All IMF level series are expressed in billions of constant 2017 international dollars at purchasing-power parity. The extension uses the July 2026 vintage of the European Commission’s AMECO database ([European Commission, 2026](#)). The relevant series are Portugal’s real GDP (OVGD), GDP deflator (PVGD), general-government gross fixed capital formation at current prices (UIGG0), and the average implicit interest rate on government debt (AYIGD). Nominal public investment is converted to a real national-currency index by deflating it with the GDP deflator:

$$I_t^{\text{AMECO,real}} = \frac{I_t^{\text{AMECO,nominal}}}{P_t^{\text{GDP}}/100}. \quad (3)$$

The level of this national-currency series is not directly combined with the IMF level. Instead, its growth rate is used to extend the IMF-consistent investment series:

$$\hat{I}_t = \hat{I}_{t-1} \left(1 + g_t^{I,\text{AMECO}} \right), \quad g_t^{I,\text{AMECO}} = \frac{I_t^{\text{AMECO,real}}}{I_{t-1}^{\text{AMECO,real}}} - 1. \quad (4)$$

This growth-rate splice preserves the IMF valuation and the level of the series at the last overlap observation while importing the subsequent AMECO dynamics. Because the IMF flow includes separately identified PPP investment whereas the AMECO extension is general-government gross

¹For broader perspectives on the composition and efficiency of Portuguese public investment, see [Pereira \(2013\)](#) and [Pereira and Pereira \(2016\)](#).

fixed capital formation, the procedure implicitly assumes that the broader IMF-consistent aggregate grows at the same rate as government investment after 2019. This is a transparent approximation rather than a direct continuation of both components. Real GDP is extended analogously with AMECO real-GDP growth:

$$\hat{Y}_t = \hat{Y}_{t-1} \left(1 + g_t^{Y, \text{AMECO}}\right). \quad (5)$$

In the source calculation, the last IMF observation is 2019 and the AMECO extension begins in 2020. In this vintage, 2025 is best regarded as an estimate and 2026–2027 as forecasts. The output series therefore mixes historical IMF estimates, AMECO-based updates for realized years, and forecast-based extensions. For this reason, all main-text time-series figures shade 2025–2027 to distinguish the forecast and estimated years from historical data.

2.2 Perpetual inventory and depreciation

The IMF timing convention assumes that new investment becomes productive halfway through the year. For the capital stock measured at the beginning of a period, the accumulation equation is

$$K_{t+1} = (1 - \delta_t)K_t + \left(1 - \frac{\delta_t}{2}\right) I_t, \quad (6)$$

where δ_t is the depreciation rate. The second term allows investment installed, on average, at midyear to undergo half a year of depreciation.

For years in which K_t , I_t , and K_{t+1} are observed, equation (6) can be inverted to back out the implied depreciation rate:

$$\delta_t = \frac{K_t + I_t - K_{t+1}}{K_t + \frac{1}{2}I_t}. \quad (7)$$

The inferred rate rises gradually in the historical file, consistent with the IMF assumption that depreciation for public assets in high-income economies increases from 2.5 percent in 1960 to 4.55 percent in 2016. The source calculation holds the last available inferred rate fixed after 2018, at approximately 4.62 percent per year. The updated stock is then obtained recursively from equation (6).

2.3 Ratios, timing, and interpretation

The reported intensity measures are

$$i_t = 100 \frac{I_t}{Y_t}, \quad k_t = 100 \frac{K_t}{Y_t}. \quad (8)$$

The stock-to-flow ratio K_t/Y_t is widely used as a scale normalization, but it should not be interpreted as a literal share of annual output. It may rise during recessions even if real capital falls, as occurred in 2020, and it can decline because GDP grows even when the real stock is approximately flat. Accordingly, the analysis reports both levels and ratios.

The estimates are gross, not efficiency-adjusted. Two economies with the same measured capital stock may obtain different services from it because of project selection, procurement, maintenance, congestion, geographic allocation, or administrative capacity. [Gupta et al. \(2014\)](#) emphasizes the distinction between measured and efficiency-adjusted public capital. Moreover, physical-capital

accounts do not measure the skills of teachers, medical personnel, engineers, or public administrators, even though these inputs complement physical assets. The calculations therefore answer a narrow but useful question: under a standardized depreciation schedule, has recorded gross fixed capital formation been sufficient to replace the public assets that wear out?

3 Results

3.1 The investment decline

Figure 1 displays real public investment from 1960 through 2027. The broad historical pattern is an extended expansion followed by a dramatic post-crisis contraction. Public investment falls especially sharply after 2010 and reaches its modern low in the middle of the 2010s. Measured relative to GDP, it declines to 1.55 percent in 2016. This compression is large enough that gross investment no longer offsets estimated depreciation.

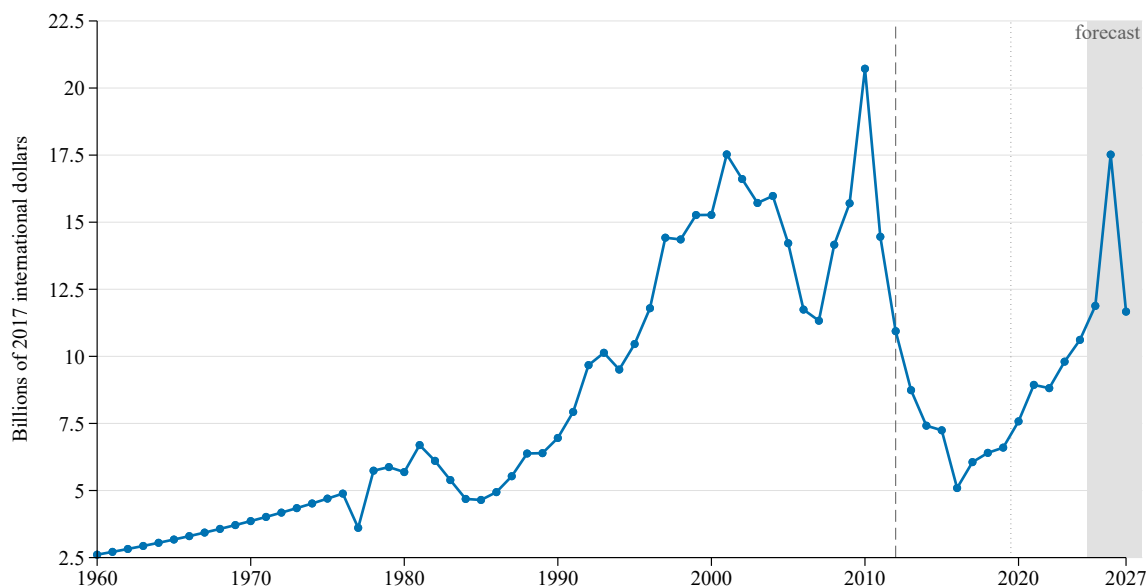


Figure 1: Real public investment in Portugal

Notes: Billions of constant 2017 international dollars. Public investment includes the public and PPP components in the IMF data and is extended after 2019 using real growth in AMECO general-government gross fixed capital formation. The dashed line marks 2012; the dotted line marks the start of the AMECO extension. Shading denotes the AMECO estimate/forecast period (2025–2027). Sources: IMF Investment and Capital Stock Dataset and AMECO, July 2026 vintage.

The update also points to a recovery, but one that must be read carefully. Investment rises to 2.71 percent of GDP in 2024 and is projected to reach 4.33 percent in 2026 before falling back to 2.83 percent in 2027. The 47 percent real growth rate embedded in the 2026 AMECO forecast is exceptional and likely reflects the timing of investment associated with European funds and the PRR. Because the spike is forecast-based and temporary, it is too early to conclude that Portugal has moved to a persistently higher public-investment regime.

Figure 2 compares public investment as a share of GDP with the average interest rate on public debt. The comparison is not an accounting identity, but it illustrates changing fiscal priorities and constraints. During much of the post-sovereign-debt-crisis period, the investment ratio remained low

even as the average interest burden declined. Lower financing costs did not automatically translate into sufficient capital formation.

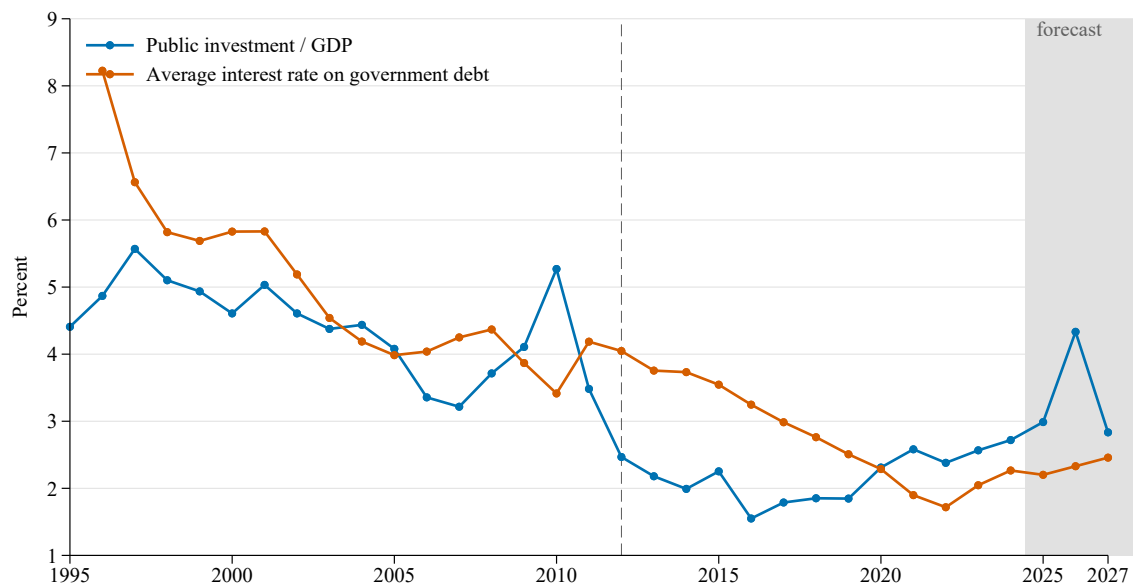


Figure 2: Public investment and the average interest rate on government debt

Notes: Percent. Both series are taken directly from AMECO. Public investment is general-government gross fixed capital formation as a share of GDP. The interest-rate series is AMECO’s average implicit interest rate on government debt. The dashed line marks 2012; shading denotes the AMECO estimate/forecast period (2025–2027). Source: AMECO, July 2026 vintage.

3.2 The composition of expenditure adjustment

AMECO’s general-government accounts make it possible to compare public investment with other expenditure categories using a common source, valuation basis, and denominator. Figure 3 reports public investment, cash social benefits, and the difference between government final consumption and public investment, all as shares of GDP. Cash social benefits correspond to AMECO’s “social benefits other than social transfers in kind” series and therefore exclude benefits provided to households in kind, which are included in government final consumption. The difference between government consumption and public investment is not a national-accounts category; it is constructed to highlight the changing gap between the two expenditure ratios.

The long-run comparison shows that the composition of spending shifted away from investment. Between 1995 and 2024, public investment fell from 4.4 to 2.7 percent of GDP. Government consumption changed comparatively little, declining from 17.5 to 16.8 percent, whereas cash social benefits rose substantially, from 10.8 to 16.0 percent. Consequently, the difference between government consumption and investment increased from 13.1 to 14.1 percent of GDP. The widening gap makes the relative compression of investment especially visible.

The same asymmetry is visible around the sovereign-debt crisis, although all categories were affected to some degree. From 2010 to 2024, public investment fell from 5.3 to 2.7 percent of GDP—a decline of almost one half. Government consumption declined from 20.6 to 16.8 percent, while cash social benefits moved only from 16.6 to 16.0 percent. The investment series also reached

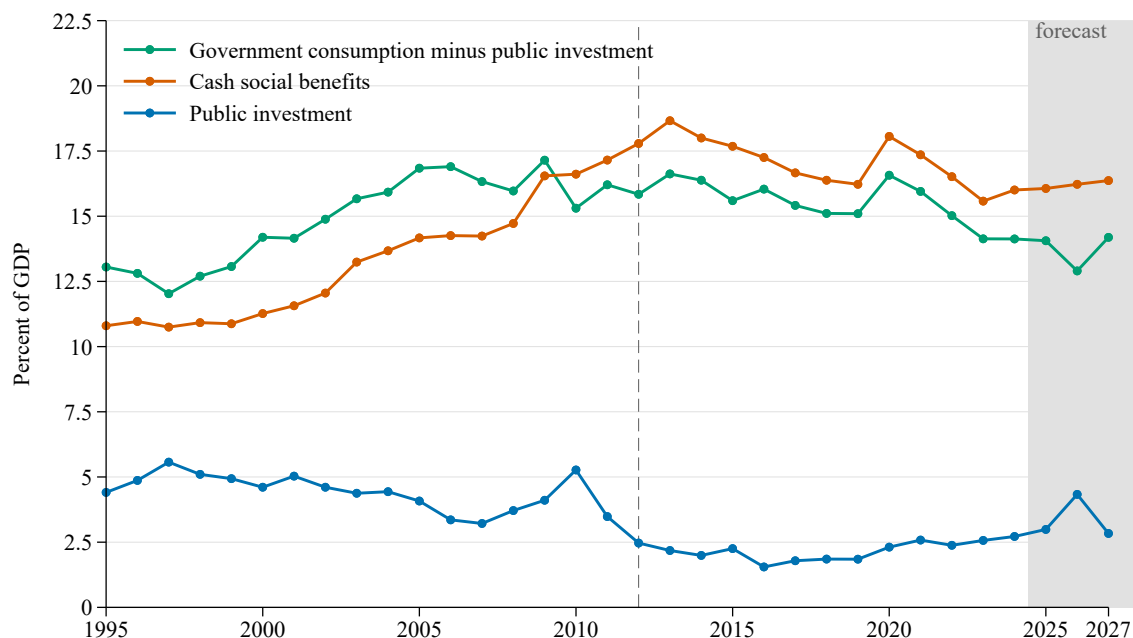


Figure 3: Public investment, government consumption less public investment, and cash social benefits in Portugal

Notes: Percent of GDP at current prices. Public investment is general-government gross fixed capital formation. “Government consumption minus public investment” is the difference between the two AMECO ratios and is not a national-accounts expenditure category. Cash social benefits exclude social transfers in kind. The dashed vertical line marks 2012. Shading denotes the AMECO estimate/forecast period (2025–2027). Source: AMECO, July 2026 vintage.

its trough much sooner, at 1.55 percent of GDP in 2016. These patterns are consistent with the political-economy mechanism discussed above: capital spending can be compressed quickly because the deterioration of the corresponding services is delayed, whereas reductions in current consumption and transfers are more immediately visible to households and public-sector employees.²

The AMECO forecast anticipates an investment surge in 2026 without a comparable change in consumption or transfers. This reinforces the importance of distinguishing a temporary, program-driven increase from a durable change in expenditure composition. A sustained rebuilding of public capital requires investment to remain sufficiently high after temporary European financing programs expire.

3.3 Accumulated erosion of the capital stock

The low-investment period translates mechanically, but gradually, into a lower stock. Figure 4 shows that real public capital peaks in 2013 at 242.6 bn 2017 international dollars. It then declines in every year through 2024, reaching 210.4 bn. The cumulative loss is 32.2 bn, or 13.3 percent of the peak stock. Put differently, by 2024 the estimated real stock had returned to approximately its level in the early 2000s, despite a substantially larger economy and greater demands on public infrastructure. A complementary recent assessment appears in Box 9 of the Banco de Portugal’s June 2026 Economic

²Alternatively, it could be argued that the decline of investment relative to other categories is a consequence of issues related to execution, i.e. the actual spending of budgeted funds. This is discussed in more detail in section 3.5.

Bulletin (Banco de Portugal, 2026). Appendix Figure A.2 plots the depreciation rate implied by the IMF stock-flow identity through 2018 and identifies the subsequent period over which the terminal rate is held fixed.

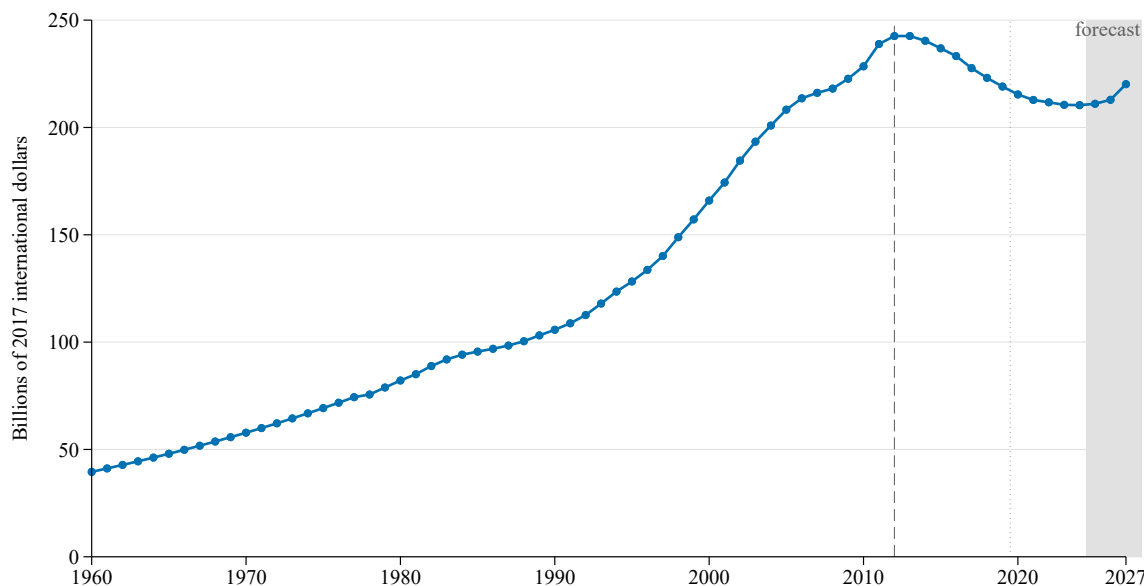


Figure 4: Estimated real public capital stock in Portugal

Notes: Billions of constant 2017 international dollars. The stock combines IMF public and PPP capital and is extended after 2019 using equation (6). The dashed vertical line marks 2012; the dotted vertical line marks the start of the AMECO extension. Shading denotes the AMECO estimate/forecast period (2025–2027). Sources: IMF and author’s calculations using AMECO, July 2026 vintage.

The timing of recovery is important. Under the IMF convention, investment during year t affects the beginning-of-year stock in $t + 1$. Public investment in 2023 was still slightly below the approximate maintenance requirement, and the 2024 increase was only just sufficient to stabilize the stock. The stock rises modestly in 2025 and 2026 and more clearly in 2027, when the projected 2026 surge is incorporated. The projected 2027 stock is 220.2 billion, about 9.2 percent below the 2013 peak. Thus even full realization of the AMECO forecast would repair only part of the accumulated decline.

Scaling by GDP makes the erosion more pronounced (Figure 5). The ratio peaks at 77.2 percent in 2013 and falls to 53.8 percent in 2024, a decline of 23.4 percentage points. The increase in 2020 is a denominator effect: real GDP fell more sharply than the capital stock during the pandemic. As growth resumed, the ratio continued downward. The projected ratio reaches 52.6 percent in 2026 and 53.4 percent in 2027, remaining close to its historical low.

3.4 Economic and policy interpretation

The accounting decline is economically relevant through two broad channels. First, public capital enters private production directly. Roads, ports, airports, energy and water networks, and digital systems reduce transport, coordination, and transaction costs. When those systems become congested, unreliable, or technologically obsolete, private firms require more time and resources to produce a given amount of output. A large empirical literature associates public capital with

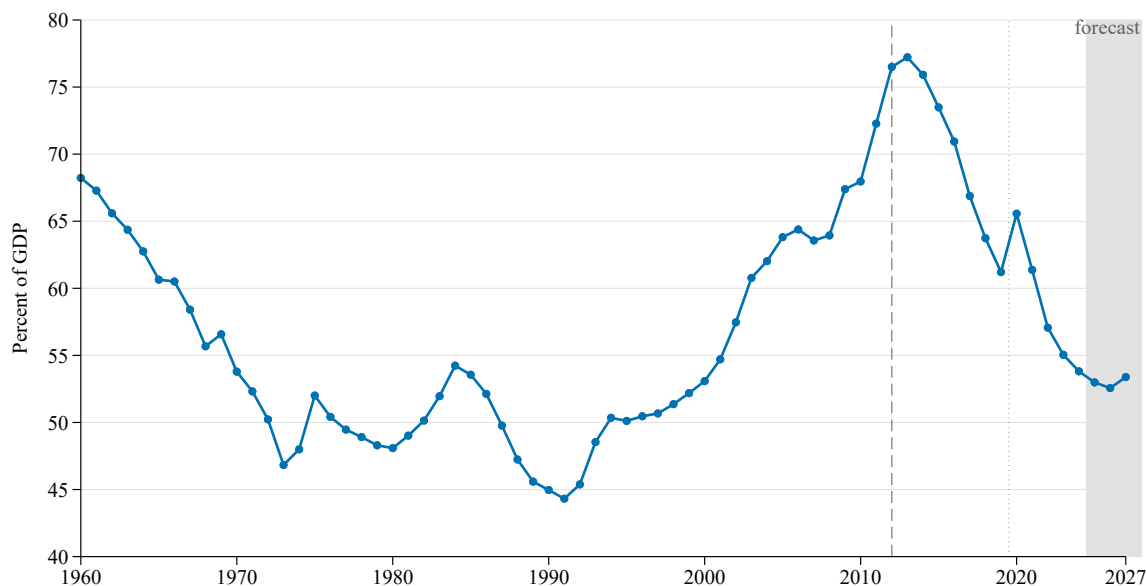


Figure 5: Estimated public capital stock relative to GDP

Notes: Percent of real GDP. Because the numerator is a stock and the denominator a flow, the ratio is a scale indicator rather than an output share. The dashed vertical line marks 2012; the dotted vertical line marks the start of the AMECO extension. Shading denotes the AMECO estimate/forecast period (2025–2027). Sources: IMF and author’s calculations using AMECO, July 2026 vintage.

Table 1: Selected observations

Year	Investment (bn 2017 international dollars)	Capital stock	Capital/GDP (percent)
2013	8.7	242.6	77.2
2019	6.6	219.1	61.2
2020	7.6	215.4	65.6
2024	10.6	210.4	53.8
2025 ^e	11.9	211.0	53.0
2026 ^e	17.5	212.9	52.6
2027 ^e	11.7	220.2	53.4

Notes: Superscript *e* denotes an AMECO estimate or projection. Investment and capital are in billions of constant 2017 international dollars. The IMF series is extended after 2019 with AMECO growth rates. Sources: IMF, AMECO July 2026 vintage, and author’s calculations.

growth and productivity, although causal returns vary across countries and projects (Pereira, 2000; Gupta et al., 2010). Section 4 further explores these effects and tries to quantify the implications of the decline in the stock of public capital for the level of GDP and growth.

Second, physical capital is complementary to labor in the provision of public services. Education depends not only on teachers but also on adequate schools, laboratories, and digital equipment. Health care depends on medical staff and organizational capacity, but also on hospitals, diagnostic equipment, and information systems. A decline in the quantity or quality of these assets can reduce current well-being and future productivity. The broader concern is therefore not that a statistical

series has fallen, but that the decline may reveal a growing mismatch between the assets available to the state and the services society expects it to provide.

These channels unfold slowly. Cutting an investment appropriation can improve the contemporaneous fiscal balance without immediately closing a road or hospital. Existing capital continues to provide services, maintenance backlogs are imperfectly observed, and users adapt. The full cost becomes visible only as failures become more frequent, capacity constraints bind, and replacement becomes unavoidable. This delay creates a political-economy asymmetry: current voters receive the benefit of fiscal savings while future taxpayers and service users bear part of the cost. Population aging may reinforce this asymmetry if electoral demand shifts toward current transfers and away from investment with long horizons.

3.5 Institutional constraints on investment execution

The amount appropriated for public investment is not necessarily the amount that is ultimately spent, and expenditure itself does not guarantee that a usable asset is delivered on time. Public investment passes through a long implementation chain: projects must be identified and appraised, procurement conducted, and contracts monitored. Failures at any stage can reduce actual investment even when financing is available. Administrative capacity is especially important when many projects are launched simultaneously, because the relevant skills—engineering, legal, procurement, and project-management expertise—are scarce within both the public administration and the firms that supply it.

Recent Portuguese evidence suggests that this distinction between budgeted and realized investment is quantitatively important. For the central-government and social-security projects classified as “structural investments,” realized expenditure fell short of the corresponding Government Budget plans by an average of EUR 719 M per year over 2019–2023. In 2024 the shortfall reached EUR 2 bn after adjusting for projects added only after the budget was published; mobility, infrastructure, and communications accounted for 42 percent of that gap (Sousa, 2025). Execution difficulties remained in 2025 despite the acceleration of investment supported by European funds. Expenditure on the investment measures covered by the Public Finance Council’s assessment was only about half the amount anticipated in the 2025 budget (Conselho das Finanças Públicas, 2026).

Low execution can therefore provide a complementary explanation for the weak investment documented above. While fiscal choices determine the amount of potential expenditure, the actual amounts that become productive capital depend on the quality of public-investment-management institutions. Thus the decline in public investment may not result solely from political choices, but may also reflect limitations in administrative and institutional capacity. Prolonged periods of low investment could themselves weaken that capacity, although the evidence presented here does not establish that institutional quality has deteriorated over time.

3.6 Investment required to preserve the 2013 capital stock

A potential useful indicator is the investment flow required to maintain the public capital stock at a given level. This indicator can be constructed directly from the perpetual-inventory equation. Applying equation (6) to a stock fixed at its 2013 peak gives

$$\tilde{I}_t^{\text{maint}} = \frac{\delta_t K_{2013}^G}{1 - \delta_t/2}. \quad (9)$$

Figure 6 compares this maintenance requirement with realized public investment, with both flows divided by realized GDP in year t .³ The required share declines gradually as GDP grows because the target real stock is fixed: it is about 3.5 percent of GDP in 2013 and 2.8 percent in 2026. Realized investment remained below the requirement for most of the intervening period. The estimated 2025 flow and especially the projected 2026 surge exceed the maintenance requirement, allowing the stock to begin recovering, although not immediately returning it to its 2013 level.

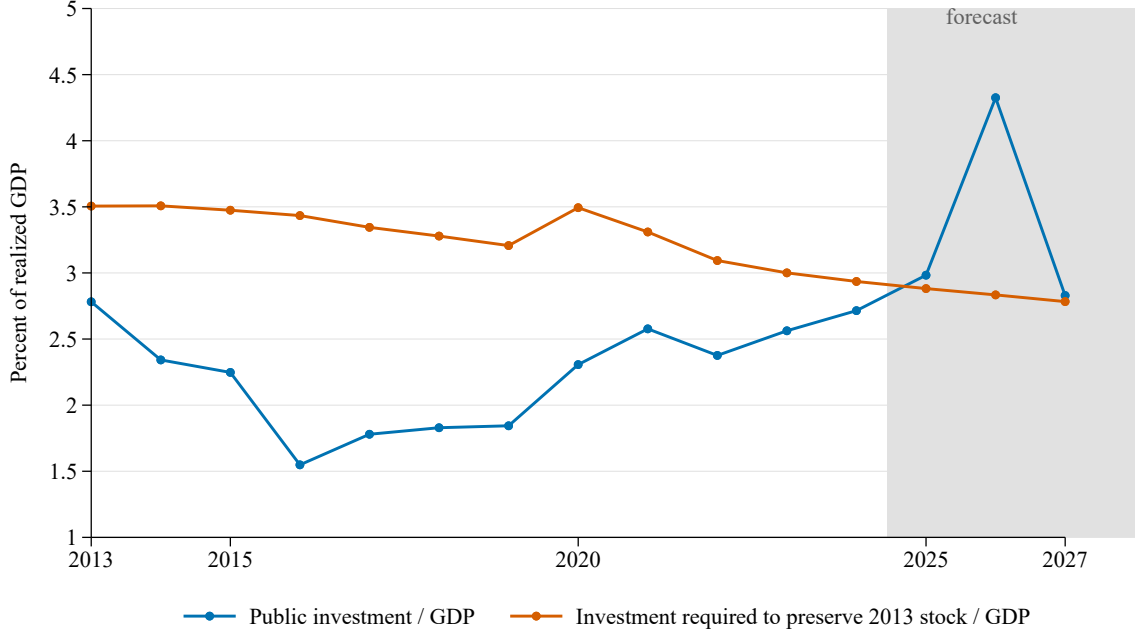


Figure 6: Public investment and the flow required to preserve the 2013 capital stock

Notes: Both series are expressed as percentages of realized GDP. The maintenance requirement is $\delta_t K_{2013}^G / (1 - \delta_t/2)$ and therefore holds the real public-capital stock fixed at its 2013 level under the IMF timing convention. Public investment is the IMF-consistent series extended with AMECO real-investment growth. Shading denotes the AMECO estimate/forecast period (2025–2027). Sources: IMF, AMECO July 2026 vintage, and author’s calculations.

4 Macroeconomic implications

Public capital can affect aggregate performance by raising the productivity of private inputs. Transport and digital networks reduce firms’ operating costs; education and health infrastructure complement workers’ skills; and reliable administrative, judicial, water, and energy systems reduce delays and uncertainty. These mechanisms suggest a positive relationship between the growth of the public capital stock and productivity growth. In this section, I first document a simple statistical association between labor-productivity and public-capital growth in Portugal. I then use a calibrated production-function framework to quantify the potential implications of the post-2013 decline in public capital for output and productivity.

³This is a simplified “partial equilibrium” measure, as GDP would have been different than its realized value given different investment policies.

4.1 Production-function framework

To organize the empirical exercise, consider an aggregate Cobb–Douglas production function that distinguishes private and public capital:

$$Y_t = A_t K_{t-1}^\alpha (K_{t-1}^G)^\eta N_t^{1-\alpha-\eta}, \quad (10)$$

where Y_t is real GDP, A_t is total factor productivity, K_{t-1} is private capital, K_{t-1}^G is public capital, and N_t is employment. Dividing by employment gives

$$\frac{Y_t}{N_t} = A_t \left(\frac{K_{t-1}}{N_t} \right)^\alpha \left(\frac{K_{t-1}^G}{N_t} \right)^\eta. \quad (11)$$

The parameter η is the direct elasticity of output with respect to public capital, holding productivity, private capital, and employment fixed.

Estimating η from a single aggregate time series is difficult. Output and both capital stocks trend strongly; public investment responds endogenously to economic and fiscal conditions; and private and public capital may be complementary. Early aggregate production-function estimates were often large, while later panel and dynamic studies generally found more moderate effects. For Portugal, [Ligthart \(2000\)](#) estimated elasticities around 0.37–0.39 using production-function, cointegration, and VAR methods. [Pina and St. Aubyn \(2005\)](#) also estimated a cointegrated production system for Portugal with private, public, and human capital. In a broad international panel, [Gupta et al. \(2010\)](#) estimated a public-capital elasticity of approximately 0.13 for OECD economies. More recent work has emphasized heterogeneous dynamics, infrastructure quality, and imperfect substitution between public and private capital ([Calderón et al., 2015](#); [An et al., 2019](#)).

Given this uncertainty, the analysis below separates two questions. First, simple time-series regressions document whether labor-productivity and public-capital growth move together in Portugal. Second, a calibrated accounting exercise applies a transparent range of external estimates, $\eta \in [0.05, 0.39]$, to the observed post-2013 capital decline. The lower end is conservative, 0.10 is used as a central calibration, and the upper end corresponds approximately to the older Portugal-specific estimates. Neither exercise identifies a causal elasticity.

4.2 Public capital and labor productivity growth

Labor productivity, LP_t , is AMECO real GDP at reference-year prices per person employed (code RVGDE). Public capital, K_t^G , is the combined public and PPP stock constructed in Section 2. I estimate the bivariate relationship

$$g_t^{LP} = \alpha + \beta_s g_{t-s}^{K^G} + u_t, \quad s \in \{0, 1, \dots, 5\}.$$

Both variables are annual log growth rates in percent. The contemporaneous specification allows productivity and public capital to move together within a year; the remaining columns allow public-capital growth to precede productivity growth by one to five years. Real-GDP growth is deliberately not included as a control because GDP is the numerator of labor productivity. The reported standard errors are heteroskedasticity-robust.

The estimated association is positive but horizon-dependent. A one-percentage-point increase

Table 2: Labor-productivity and public-capital growth

Public-capital growth dated	t	$t - 1$	$t - 2$	$t - 3$	$t - 4$	$t - 5$
Coefficient	0.4061**	0.3822**	0.3138*	0.3115	0.2527	0.2465
Robust standard error	(0.1607)	(0.1718)	(0.1772)	(0.1963)	(0.1912)	(0.1974)

Notes: Each column is a separate bivariate regression. Column headings indicate the date of public-capital growth relative to labor-productivity growth in year t . Growth rates are annual log differences in percent. Standard errors are heteroskedasticity-robust. Regressions use available observations within 1961–2024; longer lags therefore have slightly fewer observations. *, **, and *** denote significance at the 10, 5, and 1 percent levels. The sample excludes 2025–2027 estimates and forecasts. Sources: IMF, AMECO July 2026 vintage, and author’s calculations.

in contemporaneous public-capital growth is associated with approximately 0.41 percentage point higher labor-productivity growth. The one-year-lag coefficient is 0.38 and the two-year-lag coefficient is 0.31; these estimates are statistically significant at the 5, 5, and 10 percent levels, respectively. Coefficients at longer lags remain positive but are not statistically significant.

Several identification problems prevent a causal interpretation. Fiscal policy responds to the cycle, so output and investment may share common shocks; higher expected growth may itself induce public investment; completed projects may affect productivity only after long and heterogeneous lags; and the perpetual-inventory stock measures depreciated expenditure rather than the quantity or quality of capital services. Aggregation also obscures which assets matter for which sectors. A stronger causal design would require project-level or regional variation, measures of infrastructure use and quality, and instruments or institutional discontinuities that isolate investment changes unrelated to contemporaneous productivity prospects.

The β_s coefficients should therefore not be interpreted as estimates of η . Equation (11) contains private-capital intensity and unobserved productivity, while the bivariate regressions omit both. The regressions document Portuguese comovement; the calibrated exercise uses the external literature to discipline the economically relevant elasticity.

4.3 Calibrated consequences of the post-2013 decline

Consider a counterfactual in which gross investment from 2013 onward was just sufficient to hold real public capital at its 2013 peak, so that $\widetilde{K}_t^G = K_{2013}^G$. Holding A_t , private capital, and employment at their observed values, equation (10) implies

$$\ln Y_t - \ln \widetilde{Y}_t = \eta \left(\ln K_{t-1}^G - \ln \widetilde{K}_{t-1}^G \right). \quad (12)$$

Because employment is held fixed, the proportional effect on labor productivity is identical. This is a partial-equilibrium accounting calculation: it excludes crowding-in of private capital, endogenous employment and productivity responses, financing effects, and differences between measured capital and effective capital services.

The estimated stock remained below its 2013 peak through 2026. Because output in year t depends on public capital in $t - 1$, Table 3 applies alternative values of η to the decline in the stock between 2013 and 2025 to calculate the output and labor-productivity gap in 2026. The central calibration $\eta = 0.10$ translates the relevant capital shortfall into an output and labor-productivity gap and an average annual growth contribution over 2013–2026. The conservative $\eta = 0.05$ calibration yields a smaller effect, whereas the older Portugal-specific upper estimate yields

a substantially larger one. The wide range is a feature of the exercise: parameter uncertainty is economically material and should remain visible. The table shows that, depending on the value of η , the post-2013 decline in public capital could imply output 0.7–5.3 percent lower in 2026 than if the stock had been preserved at its 2013 level, equivalent to an average annual growth drag of 0.05–0.42 percentage points over 2013–2026.

Table 3: Calibrated effect of the decline in public capital

Assumed η	2026 output/prod. gap, %	Avg. ann. growth contrib., pp
0.05	-0.7	-0.05
0.10	-1.4	-0.11
0.15	-2.1	-0.16
0.20	-2.7	-0.21
0.25	-3.4	-0.27
0.30	-4.1	-0.32
0.39	-5.3	-0.42

Notes: Because output in 2026 depends on public capital in 2025, the output and labor-productivity gap is $100[(K_{2025}^G/K_{2013}^G)^\eta - 1]$. The average annual contribution is $100\eta \ln(K_{2025}^G/K_{2013}^G)/(2026 - 2013)$. Negative values indicate lower output and productivity relative to a counterfactual that holds public capital at its 2013 level. This is a calibrated partial-equilibrium accounting exercise, not a causal estimate. Sources: IMF, AMECO July 2026 vintage, and author's calculations.

Figure 7 displays the same calculation in levels, using the log of AMECO real GDP per capita. Counterfactual GDP per capita is constructed from equation (12), holding population at its observed value and public capital at its 2013 level. The central line uses $\eta = 0.10$ and the band spans 0.05–0.39. The counterfactual and actual series coincide before the capital-stock decline can affect production and then separate as the cumulative public-capital gap widens.

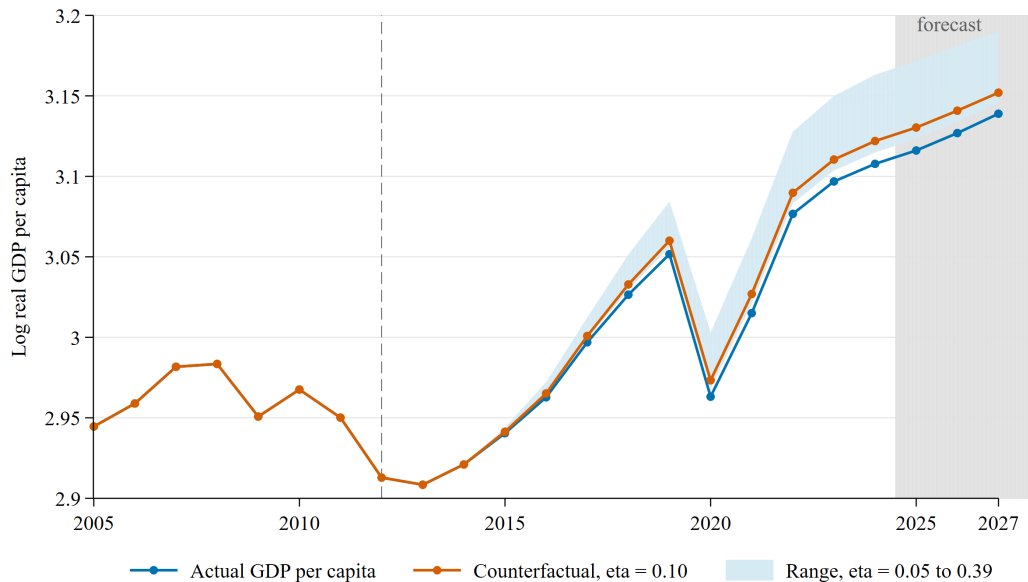


Figure 7: Actual and counterfactual real GDP per capita

Notes: Natural logarithm of real GDP per capita at 2020 reference prices (AMECO RVGDP). The counterfactual holds real public capital at its 2013 level while retaining observed population. The central counterfactual uses $\eta = 0.10$; the band spans $\eta = 0.05$ to 0.39 . Output in year t depends on public capital in $t - 1$, as in equation (10). The dashed vertical line marks 2012. Shading denotes the AMECO estimate/forecast period (2025–2027). Sources: IMF, AMECO July 2026 vintage, and author’s calculations.

The constant-stock benchmark is deliberately modest: it asks what would have happened if Portugal had merely prevented erosion, using the investment requirement documented in Section 3.6, not if public capital had continued growing at its pre-crisis rate. Two more expansionary benchmarks—continuation of a pre-2012 trend and stabilization of the capital-to-GDP ratio—would generate larger gaps but would also depend more heavily on contestable assumptions about the appropriate counterfactual path. For this reason, the constant-2013-stock case is the preferred baseline. Appendix Figures A.3 and A.4 report the corresponding GDP-per-capita counterfactuals for these two alternatives, with the ranges for the 2026 gap being 3%–21% below the counterfactual for the pre-2012 trend and 2%–14% below the counterfactual for the constant-ratio benchmark.

5 Forward-looking project scenarios

The Portuguese government has announced a substantial pipeline of transport, health, water, and housing projects for the remainder of the decade. This section uses the available project announcements and construction windows to illustrate how that pipeline could affect public investment and the public capital stock through 2035. This is just a simple scenario analysis, as published information generally identifies a total project value and a completion date but not an exact annual expenditure profile.

Table 4 groups the quantified pipeline into five categories. Railway investment includes the first two high-speed sections between Porto and Soure, the Poceirão–Bombel modernization, and the remaining Ferrovia 2020 portfolio. Urban transit covers the delayed Lisbon Metro extensions and the remaining Porto Metro expansion portfolio. The hospital category includes the Lisbon Oriental,

Algarve Central, and Seixal projects, while water infrastructure covers the Algarve desalination plant and the Pisão–Crato and Pomarão projects. Public housing is shown separately because it is a decentralized program rather than a single infrastructure project. The resulting investment portfolio corresponds to total expenditure of about EUR 11.3 bn, of which only the shares assigned to 2028 and later are added to the scenarios. Appendix Table A.1 reports the individual projects, assumed amounts, windows, amount basis, and accounting treatment. The underlying information comes from Infraestruturas de Portugal, government announcements, and the Public Finance Council (Infraestruturas de Portugal, 2026a,b,c; Governo da República Portuguesa, 2025a,b, 2026b,c; Sousa, 2025).

Table 4: Public-investment pipeline used in the scenarios

Type	Total (EUR bn)	Approx. window	Treatment	Source
Railway	3.75	2025–2032	Construction/public works only	IP; CFP
Urban transit	1.66	2024–2029	Residuals mechanically imputed	Government; CFP
Hospitals	1.29	2026–2030	Construction-equivalent envelopes	Government; CFP
Water infrastructure	0.41	2027–2030	Approximate; timing imputed	Government; PT2030
Public housing	4.20	2024–2030	Decentralized program	Government
Total	11.31	2024–2032		

Notes: Amounts are announced or estimated nominal envelopes in current euros and are not directly summed into the IMF-valued real-investment series. They are allocated in nominal terms and deflated with the contemporaneous GDP deflator. For Oiã–Soure, the table includes only the EUR 600 M ceiling for associated public works and excludes the EUR 1.603 bn net-present-value ceiling for the PPP concession, which also covers maintenance and availability. Windows combine published dates with transparent timing assumptions where annual profiles are unavailable. Only mechanically imputed expenditure assigned to 2028 or later is added to the baseline. The new Lisbon airport, the Third Tagus Bridge, and later high-speed sections are excluded from the first three scenarios and enter only the speculative extension documented in Table 5. Appendix Table A.1 provides project-level assumptions and sources for the announced pipeline.

I consider four scenarios, expressed first in nominal euros. Let P_t denote the GDP deflator, normalized to 100 in 2020, and let $\bar{I}_{20:24}^R$ denote average real public investment in euros over 2020–2024. After the final AMECO observation, I assume that the deflator grows at 2 percent per year, so $P_t = P_{2027}(1.02)^{t-2027}$. Nominal investment in the *baseline* is

$$I_t^B = \bar{I}_{20:24}^R \frac{P_t}{100}. \quad (13)$$

Thus nominal baseline investment grows with inflation while its real value remains constant at the 2020–2024 average. This benchmark avoids extrapolating the unusually high investment projected for 2026–2027.

In the *announced-project* scenario, each reported nominal envelope N_j is allocated across its

execution window using weights $\omega_{t,j}$:

$$I_t^P = I_t^B + \sum_j \omega_{t,j} N_j. \quad (14)$$

I use S-shaped profiles that place the largest shares near the middle of each construction window; Appendix Section A.1 provides a five-year example. For projects already under way, I retain only the mechanically allocated post-2027 expenditure. In the *execution-risk* scenario, 75 percent of the scheduled envelope is realized with a one-year delay:

$$I_t^E = I_t^B + 0.75 \sum_j \omega_{t-1,j} N_j. \quad (15)$$

This parsimonious haircut captures both under-execution and potential delay.

The fourth, *speculative*, scenario adds projects with a significant political commitment but with uncertain costs and schedules as of this writing. It adds the midpoint of the published cost range for the Luís de Camões Airport, a preliminary estimate for the Third Tagus Bridge, the Soure–Carregado high-speed section, and two later Porto–Valença sections to the announced-project path:

$$I_t^S = I_t^P + \sum_m \nu_{t,m} N_m^S. \quad (16)$$

Here N_m^S denotes a speculative fixed nominal envelope and ν_{mt} its timing weight. The airport is expected to be financed primarily by the airport concessionaire rather than directly through the state budget. Consequently, this path should be interpreted as a broad public-infrastructure-equivalent scenario, consistent with the paper’s inclusion of PPP capital, rather than a forecast of general-government gross fixed capital formation. Table 5 gives the assumptions and sources (Governo da República Portuguesa, 2026a; SIC Notícias, 2026; Executive Digest, 2025; Jornal de Negócios and Lusa, 2025; SIC Notícias and Lusa, 2024).

All four paths in equations (13)–(16) are nominal. I divide each by $P_t/100$ to obtain real euros and apply its proportional deviation from the real-euro baseline to the IMF-valued investment series. This ratio splice avoids imposing a PPP level conversion. The resulting real investment paths feed into equation (6): investment in year t raises K_{t+1} by $(1 - \delta/2)I_t$, while the existing stock depreciates at 4.62 percent, the assumed constant value from 2018 onward. Real GDP grows after 2027 at its AMECO 2027 growth rate, approximately 1.82 percent per year.

This treatment is deliberately stylized. A fixed reported nominal envelope purchases fewer real resources when its expenditure occurs later and is assumed not to be subsequently indexed. Contractual indexation, changes in scope, cost overruns, or construction-cost inflation different from the GDP deflator would imply a different real path.

Figure 8 summarizes the results. The 2020–2024 average is about 9.1 billion in the IMF’s constant-price units. The announced pipeline raises real public investment to a peak of 12.4 billion in 2028; the execution-risk path peaks at 11.6 billion in 2029. Relative to GDP, those peaks are 3.0 and 2.7 percent, respectively, compared with 2.2 percent in the 2028 baseline. The speculative extension peaks at 14.9 billion in 2030, or 3.4 percent of GDP. By 2035, the announced-project scenario produces a public capital stock of about 219.0 billion, compared with 213.5 billion in the baseline. The execution-risk stock reaches approximately 217.7 billion, while the speculative stock reaches 235.7 billion. These additions slow, but do not reverse, the decline in the capital-to-GDP

Table 5: Additional assumptions for the speculative scenario

Type	Project or program	EUR m	Start	End	Basis and treatment
Airport	Luís de Camões Airport	8,400	2029	2033	Midpoint of EUR 7.9–8.9 bn range at 2024 prices; Concession financed; treated as a broad infrastructure equivalent (Government)
Bridge	Third Tagus Bridge (Chelas–Barreiro)	3,000	2029	2034	Publicly reported preliminary estimate; Road and rail construction; amount and timing remain highly uncertain (Executive Digest)
High-speed rail	Soure–Carregado	2,000	2028	2032	Environmental-study construction estimate; Four-year works; timing aligned with announced 2032 completion (Jornal de Negócios/Lusa)
High-speed rail	Porto Airport–Braga–Valença	4,000	2028	2033	Two later sections at roughly EUR 2 bn each; Construction-envelope proxy; scope and timing remain preliminary (SIC Notícias/Lusa)
Total		17,400			

Notes: Amounts are treated as fixed nominal envelopes in current euros. The airport midpoint was reported at 2024 prices but is treated as a fixed nominal total for comparability; this assumption does not incorporate subsequent contractual indexation. The Third Tagus Bridge amount is a preliminary press estimate. The Porto–Valença entry combines two sections reported at approximately EUR 2 bn each. Sources: Portuguese Government, SIC Notícias/Lusa, Jornal de Negócios/Lusa, and Executive Digest ([Governo da República Portuguesa, 2026a](#); [SIC Notícias, 2026](#); [Executive Digest, 2025](#); [Jornal de Negócios and Lusa, 2025](#); [SIC Notícias and Lusa, 2024](#)).

ratio generated by the assumption that GDP continues to grow after the project pipeline is completed. In 2035 the ratio is 46.0 percent under announced projects, 45.7 percent under execution risk, and 49.5 percent in the speculative scenario, compared with 44.8 percent in the baseline.

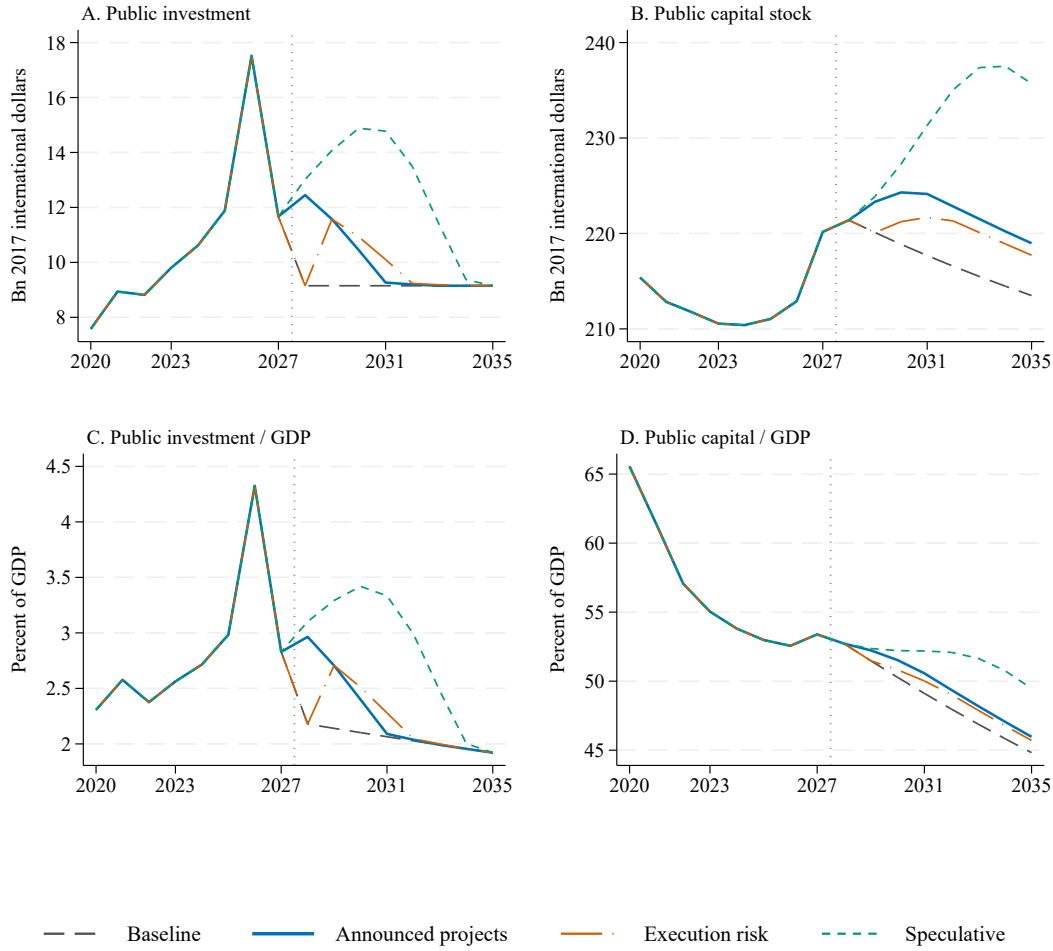


Figure 8: Public investment and public capital under alternative project scenarios

Notes: All scenarios coincide through 2027. Thereafter nominal baseline investment grows with the GDP deflator, assumed to rise by 2 percent annually, so real baseline investment remains at its 2020–2024 average. Announced projects add mechanically allocated shares of fixed nominal envelopes; execution risk realizes 75 percent with a one-year delay. The speculative scenario adds the projects in Table 5. Public capital follows equation (6), including the $(1 - \delta/2)$ adjustment, so investment in t enters the reported stock in $t + 1$. Real GDP grows at 1.82 percent per year after 2027. Sources: AMECO July 2026 vintage, sources listed in Tables 4, A.1, and 5, and author’s calculations.

6 Conclusion

Portugal’s post-2010 reduction in public investment was not merely a temporary decline in a spending flow: it generated a persistent contraction in the public capital stock. The updated series indicates that real public capital fell by 13.3 percent between its 2013 peak and 2024, while the capital-to-GDP ratio declined by more than 23 percentage points. This erosion is consistent with a decade in which gross investment was frequently insufficient to offset depreciation, even as government consumption and cash social benefits proved considerably more resilient as shares of GDP.

The July 2026 AMECO vintage contains encouraging but provisional news. It projects a very

large increase in investment in 2026 and a partial recovery of the stock in 2027. This projection should be neither dismissed nor mistaken for an accomplished reversal. The projected increase is unusually large and temporary, and even the implied 2027 stock remains materially below its 2013 peak. Sustained rebuilding would require investment to remain above the maintenance threshold beyond the completion of the PRR and other temporary programs.

The macroeconomic exercise provides a quantitative interpretation of this accounting decline. The descriptive regressions indicate positive co-movement between public-capital and labor-productivity growth, but they do not identify a causal effect. I therefore complement them with a calibrated production-function framework that applies a range of elasticities from the literature to the observed post-2013 decline in public capital. Depending on the assumed elasticity, the results imply that output and labor productivity in 2026 could be 0.7–5.3 percent lower than under a counterfactual that preserves the 2013 stock. This corresponds to an average annual growth drag of 0.05–0.42 percentage points over 2013–2026. These calculations remain partial-equilibrium exercises. A fuller evaluation of the macroeconomic consequences of public underinvestment would require a structural model that accounts for the endogeneity of investment, complementarities between public and private capital, and the dynamic responses of employment and productivity.

An important caveat is that the analysis focuses on a narrow measure of public capital: a perpetual-inventory estimate of physical assets constructed from recorded gross fixed capital formation. It does not capture human capital, the quality of public services, or the efficiency of public administration. The decline in the measured stock may therefore understate the broader erosion of productive capacity and service quality that can accompany prolonged underinvestment.

Public-investment decisions shape both the productive capacity of the state and macroeconomic performance over horizons far longer than an annual budget or electoral term. Portugal’s estimated public capital stock remains substantially below its peak, and a short-lived investment surge cannot by itself undo more than a decade of erosion. Preserving the stock requires a durable institutional commitment to maintenance and replacement. My interpretation is that the secular decline in public investment after 2010 may represent more than a temporary fiscal adjustment: it may reflect a structural shift in the political economy of public spending. Demographic forces, including population aging and declining fertility, may contribute to such a shift by increasing pressure for current expenditure relative to investment with long-dated returns. This interpretation is necessarily conjectural. Nevertheless, absent a durable change in investment priorities, the long-term outlook for Portugal’s public capital stock remains concerning.

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A Project-level assumptions

Table A.1: Detailed project assumptions for the forward-looking scenarios

Type	Project or program	EUR m	Start	End	Amount basis	Treatment
Railway	High-speed rail Porto-Oiã	2,080	2026	2030	Full construction-envelope estimate	Excludes long-run availability payments; annual timing imputed (IP)
Railway	High-speed rail Oiã-Soure	600	2027	2032	Associated public-works ceiling	Excludes the EUR 1.603 bn PPP concession NPV because it includes maintenance and availability services (IP)
Railway	Poçoirão-Bombel modernization	270	2026	2029	Contracted construction cost	Construction only (IP)
Railway	Remaining Ferrovias 2020 portfolio	801	2025	2030	Outstanding portfolio after 2024	Timing imputed (CFP)
Urban transit	Lisbon Metro Red Line	405	2024	2028	Full program envelope	One-year spillover beyond latest official target assumed (Government)
Urban transit	Lisbon Metro Violet Line	527	2026	2029	Full program envelope	Two-year spillover beyond latest official target assumed (Government)
Urban transit	Porto Metro expansion portfolio	732	2025	2028	Outstanding portfolio after 2024	Timing imputed (CFP)
Hospitals	Lisbon Oriental Hospital	796	2026	2030	Construction-equivalent envelope	Maintenance and operating payments excluded; annual timing imputed (Government)
Hospitals	Algarve Central Hospital	427	2028	2030	Construction-envelope estimate	Construction window from tender announcement (Government)
Hospitals	Seixal Hospital	67	2027	2030	Full project value	Timing imputed (CFP)
Water infrastructure	Algarve desalination plant	110	2027	2030	Approximate project envelope	Timing imputed (Government)
Water infrastructure	Pisões-Cratos hydraulic project	230	2027	2030	Approximate project envelope	Timing imputed (Government)
Water infrastructure	Pomarão water transfer	70	2027	2030	Approximate project envelope	Timing imputed (Government)
Public housing	Public and affordable housing program	4,200	2024	2030	Full decentralized-program envelope	Annual timing imputed; not a single project (Government)

Notes: Amounts are current-euro project or program envelopes, rounded to the nearest million. The amount-basis column distinguishes full envelopes, outstanding portfolios, and other estimates. Start and end dates are construction or execution assumptions. For projects already underway, the displayed envelope is allocated mechanically over the assumed window and only its imputed share from 2028 onward enters the scenario. IP denotes Infraestruturas de Portugal; CFP denotes Conselho das Finanças Públicas. The airport at Alcochete, Third Tagus Bridge, and later high-speed sections are reported separately in Table 5. See Section 5 for the conversion, timing, and execution assumptions.

A.1 Illustrative S-shaped expenditure profile

For most projects, detailed annual data on actual or planned disbursements is not available. In this case, the scenarios allocate each envelope over its assumed execution window using a stylized profile. Expenditure starts relatively slowly while design, permitting, mobilization, and early works take place; it rises as construction reaches full scale; and it declines as the project approaches completion. Thus annual expenditure is hump-shaped and cumulative expenditure follows an S-shaped path. For a representative five-year project, the assumed annual weights are 10, 20, 30, 25, and 15 percent. These weights sum to one and imply cumulative execution of 10, 30, 60, 85, and 100 percent, as illustrated in Figure A.1. Projects with other durations use the analogous profiles implemented in the scenario calculation; these are timing assumptions rather than project-specific forecasts.

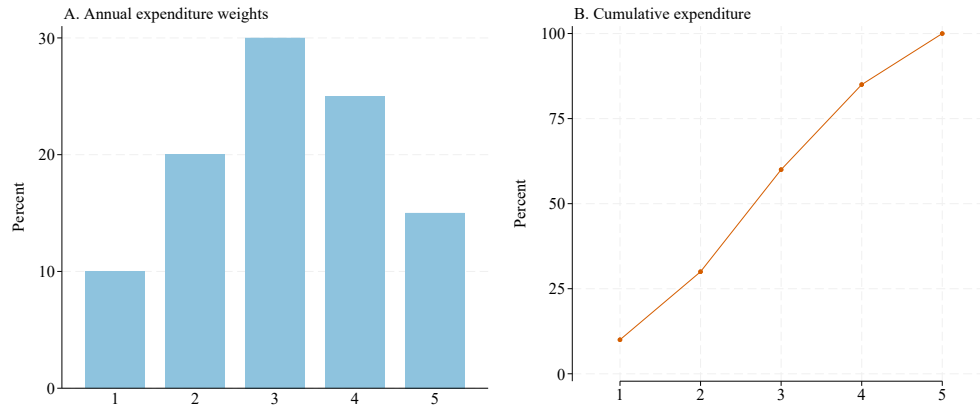


Figure A.1: Illustrative expenditure profile for a five-year project

Notes: Panel A reports the share of the project envelope assigned to each execution year. Panel B reports the cumulative share. The assumed annual weights are 10, 20, 30, 25, and 15 percent.

B Additional figures

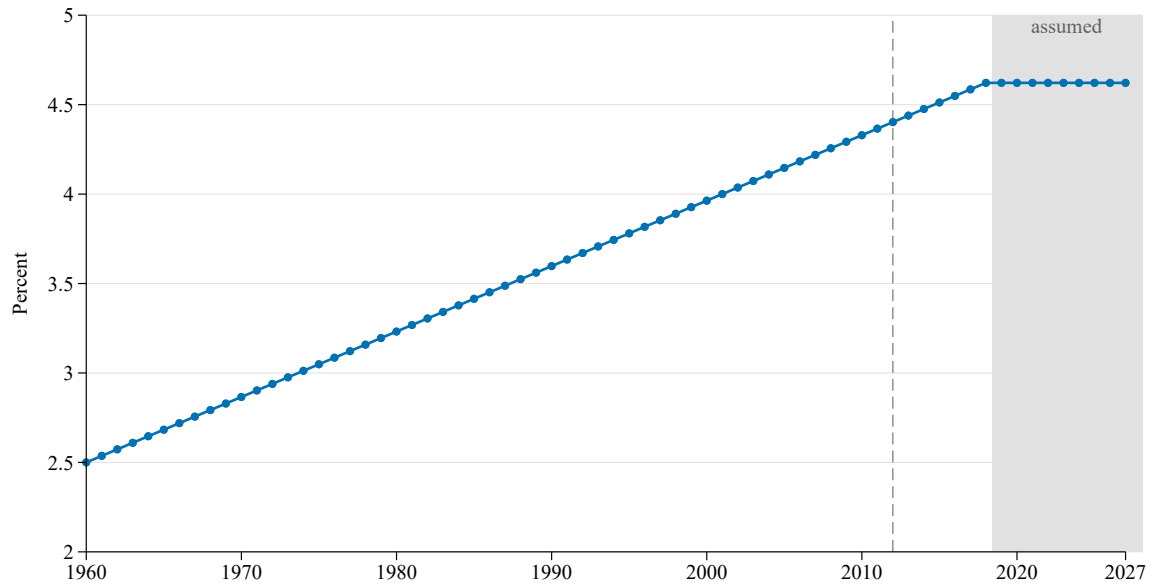


Figure A.2: Implied and assumed depreciation rate

Notes: Percent per year. Through 2018, the depreciation rate is inferred from the IMF stock-flow identity in equation (7). From 2019 onward, it is held fixed at its 2018 value of approximately 4.62 percent. The shaded region identifies the period in which the rate is assumed rather than inferred. The dashed vertical line marks 2012. Sources: IMF and author's calculations.

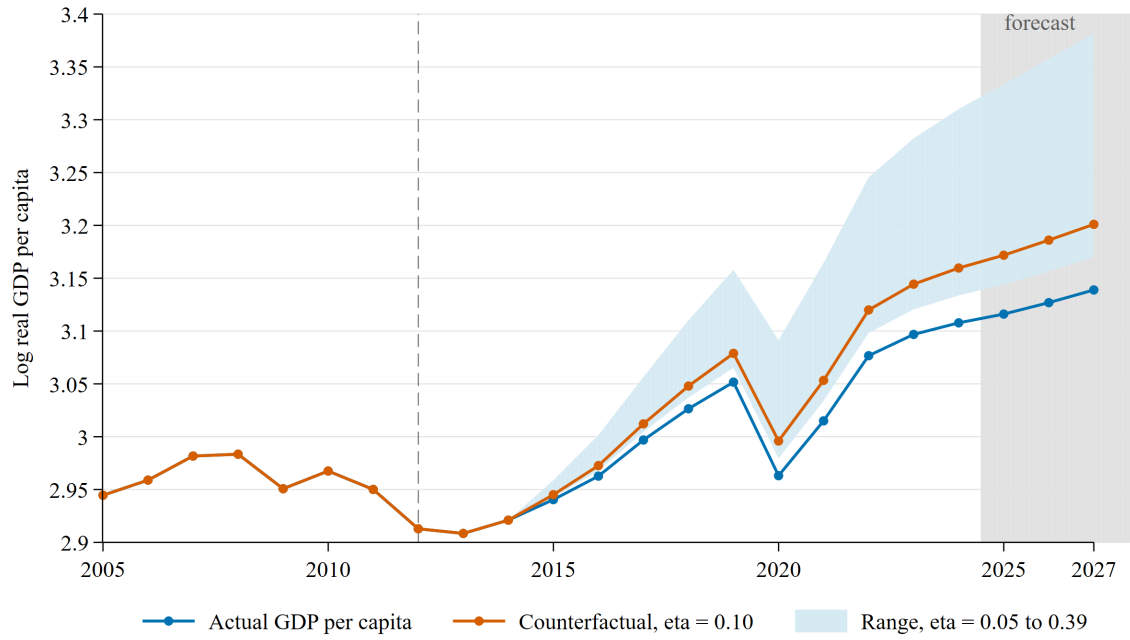


Figure A.3: GDP per capita under a continuation of the pre-2012 public-capital trend

Notes: Natural logarithm of real GDP per capita at 2020 reference prices (AMECO RVGDP). The counterfactual public-capital path extrapolates the log-linear trend estimated over 1995–2012 and is anchored at the observed 2013 stock. Population follows its observed path. The central counterfactual uses $\eta = 0.10$; the band spans $\eta = 0.05$ to 0.39 . Output in year t depends on public capital in $t - 1$. The dashed vertical line marks 2012. Shading denotes the AMECO estimate/forecast period (2025–2027). Sources: IMF, AMECO July 2026 vintage, and author's calculations.

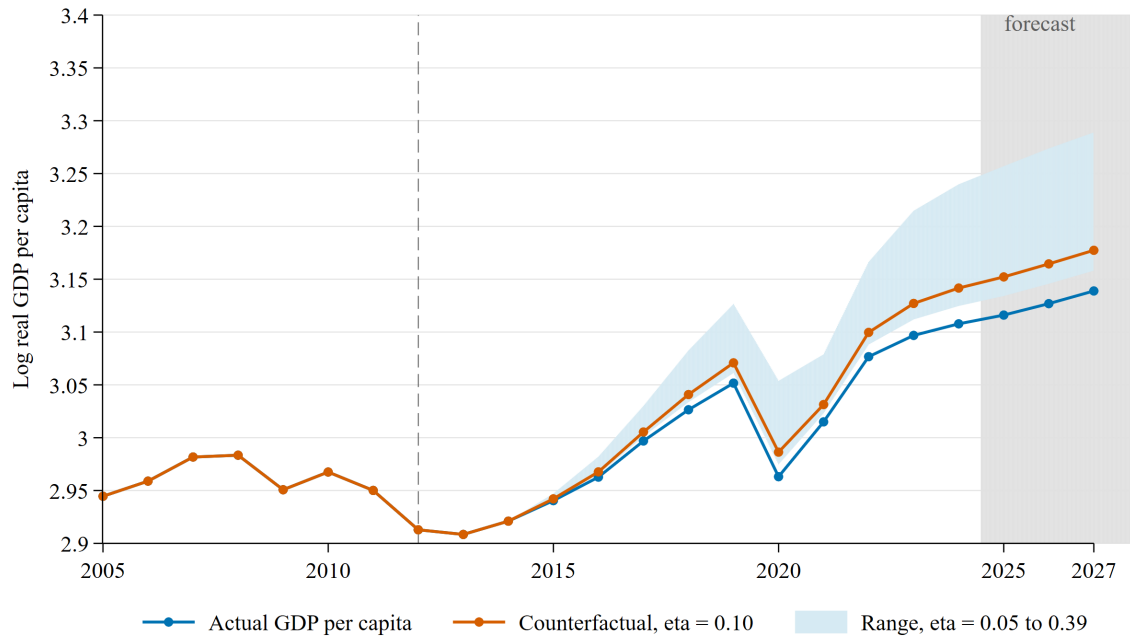


Figure A.4: GDP per capita with the public-capital-to-GDP ratio fixed at its 2013 level

Notes: Natural logarithm of real GDP per capita at 2020 reference prices (AMECO RVGDP). Counterfactual public capital equals 77.2 percent of realized GDP, its 2013 ratio, while population follows its observed path. The central counterfactual uses $\eta = 0.10$; the band spans $\eta = 0.05$ to 0.39 . Output in year t depends on public capital in $t - 1$. The dashed vertical line marks 2012. Shading denotes the AMECO estimate/forecast period (2025–2027). Sources: IMF, AMECO July 2026 vintage, and author's calculations.